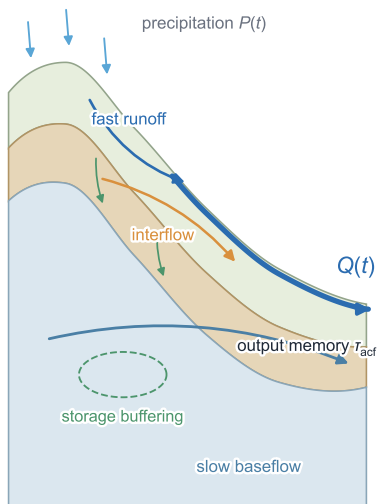
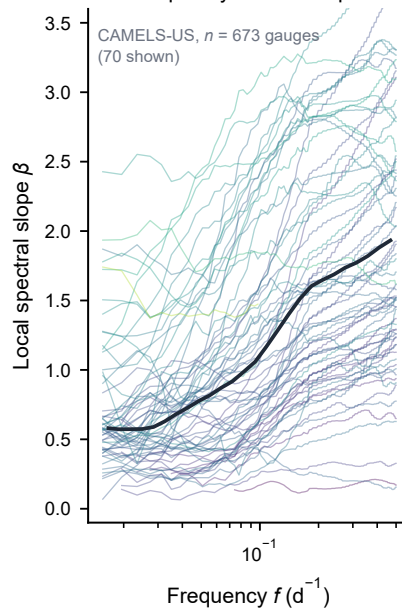


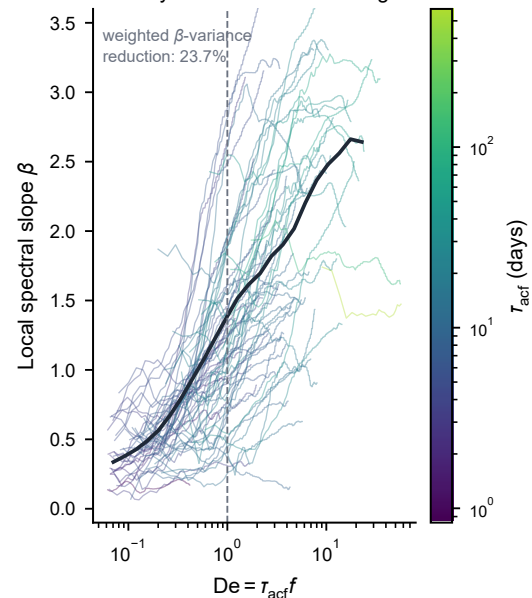
(a) Output-memory mechanism (schematic)



(b) Raw frequency: curves disperse



(c) Memory coordinate: curves align



(d) Inference pipeline and claim boundary



**Supported** cross-archive coordinate utility; model-output diagnostic beyond NSE/KGE

**Not supported** groundwater residence time; tracer causality; excess over the matched AR(1) floor